

LAB 1. ENVIRONMENT SETUP & JAVA BASICS

IT3103E – 162065

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2.2.1

- Code:

```
package lab01;
public class HelloWorld {

    public static void main(String args[]) {
        System.out.println("Xin chao \n cac ban!");
        System.out.println("Hello \t world!");
    }
}
```

- Result:

```
Xin chao
  cac ban!
Hello    world!
```

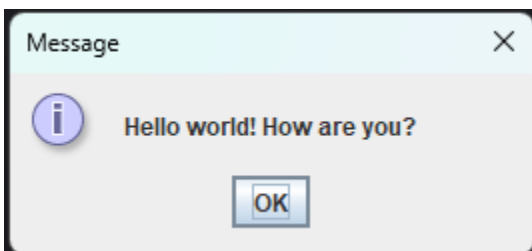
2.2.2

- Code:

```
package lab01;
import javax.swing.JOptionPane;

public class FirstDialog {
    public static void main(String[] args) {
        JOptionPane.showMessageDialog(null, "Hello world! How are you?");
        System.exit(0);
    }
}
```

- Result:



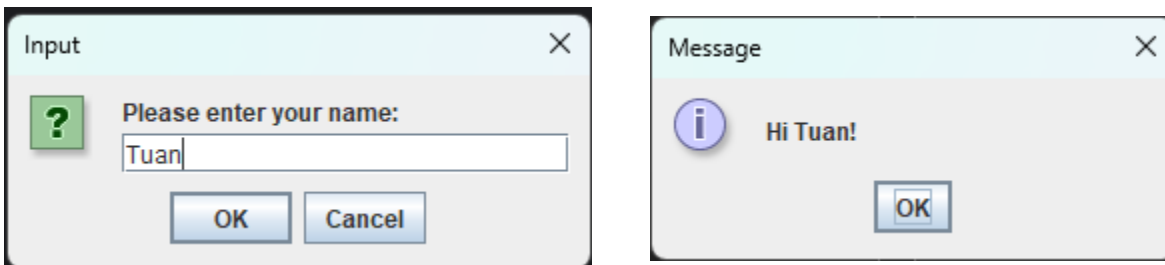
2.2.3

- Code:

```
package lab01;
import javax.swing.JOptionPane;

public class HelloNameDialog {
    public static void main(String[] args) {
        String result;
        result = JOptionPane.showInputDialog("Please enter your name:");
        JOptionPane.showMessageDialog(null, "Hi " + result + "!");
        System.exit(0);
    }
}
```

- Result:



2.2.4

- Code:

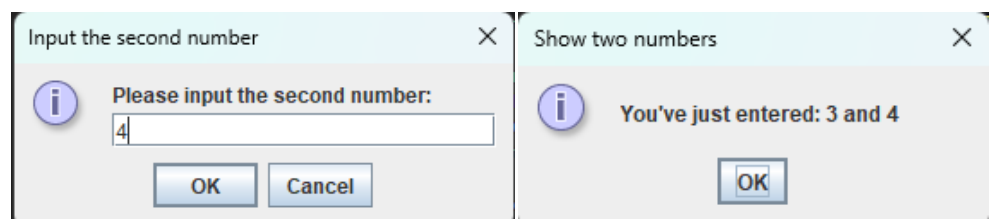
```
package lab01;
import javax.swing.JOptionPane;
public class ShowTwoNumbers {
    public static void main(String[] args){
        String strNum1, strNum2;
        String strNotification = "You've just entered: ";

        strNum1 = JOptionPane.showInputDialog(null,
            "Please input the first number: ", "Input the first number",
            JOptionPane.INFORMATION_MESSAGE);
        strNotification += strNum1 + " and ";

        strNum2 = JOptionPane.showInputDialog(null,
            "Please input the second number: ", "Input the second number",
            JOptionPane.INFORMATION_MESSAGE);
        strNotification += strNum2;

        JOptionPane.showMessageDialog(null, strNotification,
            "Show two numbers", JOptionPane.INFORMATION_MESSAGE);
        System.exit(0);
    }
}
```

- Result:



Input the first number

Please input the first number:

3

OK Cancel

2.2.5

- Code:

```

package lab01;
import javax.swing.JOptionPane;

public class MathCalc {
    public static void main(String[] args){
        String strNum1, strNum2;
        String strNotification = "Result:\n";

        // Input first number
        strNum1 = JOptionPane.showInputDialog(null,
            "Please input the first number:", "Input the first number",
            JOptionPane.INFORMATION_MESSAGE);
        double num1 = Double.parseDouble(strNum1);

        // Input second number
        strNum2 = JOptionPane.showInputDialog(null,
            "Please input the second number:", "Input the second number",
            JOptionPane.INFORMATION_MESSAGE);
        double num2 = Double.parseDouble(strNum2);

        // Perform calculations
        strNotification += "Sum: " + (num1 + num2) +
            "\nDiff: " + Math.abs(num1 - num2);

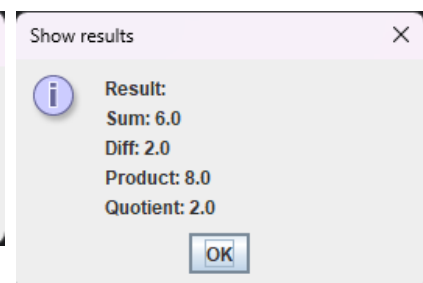
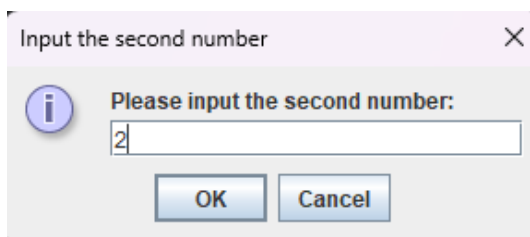
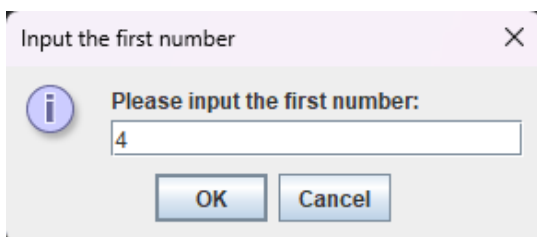
        if (num2 == 0) {
            strNotification += "\nProduct: " + (num1 * num2) +
                "\nCan't divide by zero.";
        } else {
            strNotification += "\nProduct: " + (num1 * num2) +
                "\nQuotient: " + (num1 / num2);
        }

        // Show result
        JOptionPane.showMessageDialog(null, strNotification,
            "Show results", JOptionPane.INFORMATION_MESSAGE);
        System.exit(0);
    }
}

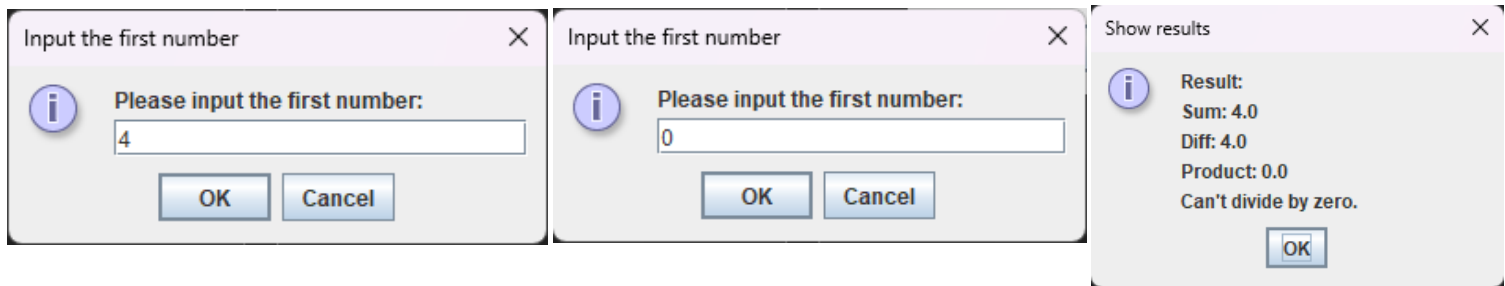
```

- Result:

+ Case 1: The second number is not 0



+ Case 2: The second number is 0



2.2.6

- Code:

+ Choosing options on the dialog to choose the program:

```
package lab01;
import javax.swing.JOptionPane;

public class SolveEquation {
    public static void main(String[] args) {
        while (true) {
            String[] options = {
                "Solve linear equation (1 variable)",
                "Solve system of 2 linear equations (2 variables)",
                "Solve quadratic equation (1 variable)",
                "Exit"
            };

            int choice = JOptionPane.showOptionDialog(
                null,
                "Choose the type of equation you want to solve:",
                "Equation Solver",
                JOptionPane.DEFAULT_OPTION,
                JOptionPane.INFORMATION_MESSAGE,
                null,
                options,
                options[0]
            );

            if (choice == -1 || choice == 3) {
                JOptionPane.showMessageDialog(null, "Goodbye!");
                break;
            }

            switch (choice) {
                case 0:
                    JOptionPane.showMessageDialog(null, "Solve Linear equation: ax + b = 0");
                    solveLinearEquationOneVar();
                    break;
                case 1:
                    JOptionPane.showMessageDialog(null, "Solve System of 2 linear equations: a11*x + a12*y = b1   a21*x + a22*y = b2");
                    solveSystemOfTwoLinearEquations();
                    break;
                case 2:
                    JOptionPane.showMessageDialog(null, "Solve Quadratic Equation: a*2*x + bx + c = 0");
                    solveQuadraticEquation();
                    break;
            }
        }
    }
}
```

+ Solving Linear

equation:

```
private static void solveLinearEquationOneVar() {
    double a = Double.parseDouble(JOptionPane.showInputDialog("Enter coefficient a:"));
    double b = Double.parseDouble(JOptionPane.showInputDialog("Enter coefficient b:"));

    if (a == 0) {
        if (b == 0)
            JOptionPane.showMessageDialog(null, "Infinite solutions.");
        else
            JOptionPane.showMessageDialog(null, "No solution.");
    } else {
        double x = -b / a;
        JOptionPane.showMessageDialog(null, "Solution: x = " + x);
    }
}
```

+ Solving System of Two linear equations:

```
private static void solveSystemOfTwoLinearEquations() {
    double a11 = Double.parseDouble(JOptionPane.showInputDialog("Enter a11:"));
    double a12 = Double.parseDouble(JOptionPane.showInputDialog("Enter a12:"));
    double b1 = Double.parseDouble(JOptionPane.showInputDialog("Enter b1:"));
    double a21 = Double.parseDouble(JOptionPane.showInputDialog("Enter a21:"));
    double a22 = Double.parseDouble(JOptionPane.showInputDialog("Enter a22:"));
    double b2 = Double.parseDouble(JOptionPane.showInputDialog("Enter b2:"));

    double D = a11 * a22 - a21 * a12;
    double D1 = b1 * a22 - b2 * a12;
    double D2 = a11 * b2 - a21 * b1;

    if (D == 0) {
        if (D1 == 0 && D2 == 0)
            JOptionPane.showMessageDialog(null, "Infinite solutions.");
        else
            JOptionPane.showMessageDialog(null, "No solution.");
    } else {
        double x = D1 / D;
        double y = D2 / D;
        JOptionPane.showMessageDialog(null, "Solution: x = " + x + ", y = " + y);
    }
}
```

+ Solving Quadratic equation:

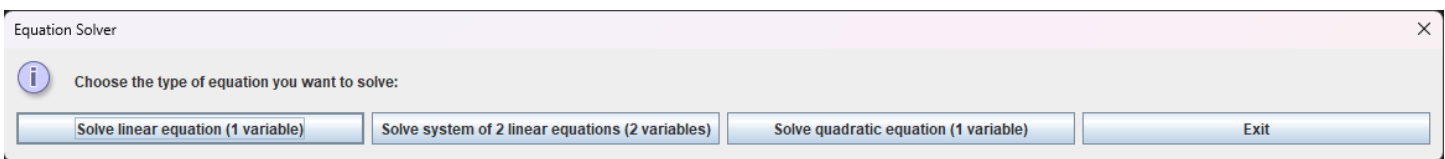
```
private static void solveQuadraticEquation() {
    double a = Double.parseDouble(JOptionPane.showInputDialog("Enter a:"));
    double b = Double.parseDouble(JOptionPane.showInputDialog("Enter b:"));
    double c = Double.parseDouble(JOptionPane.showInputDialog("Enter c:"));

    if (a == 0) {
        if (b == 0)
            JOptionPane.showMessageDialog(null, (c == 0) ? "Infinite solutions." : "No solution.");
        else {
            double x = -c / b;
            JOptionPane.showMessageDialog(null, "Linear case: x = " + x);
        }
        return;
    }

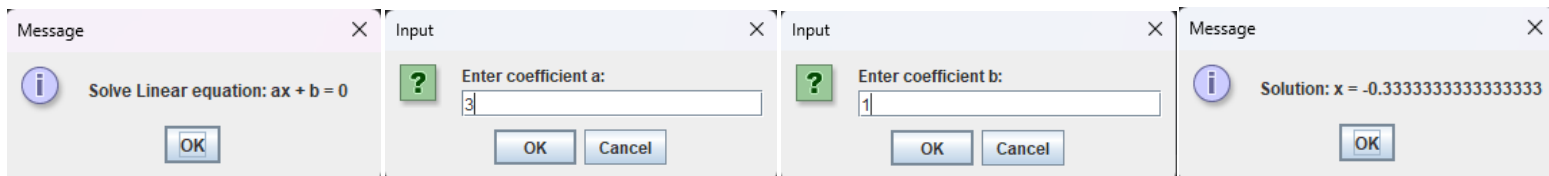
    double delta = b * b - 4 * a * c;

    if (delta < 0)
        JOptionPane.showMessageDialog(null, "No real solutions.");
    else if (delta == 0) {
        double x = -b / (2 * a);
        JOptionPane.showMessageDialog(null, "Double root: x = " + x);
    } else {
        double x1 = (-b + Math.sqrt(delta)) / (2 * a);
        double x2 = (-b - Math.sqrt(delta)) / (2 * a);
        JOptionPane.showMessageDialog(null, "Two distinct roots:\n x1 = " + x1 + "\n x2 = " + x2);
    }
}
```

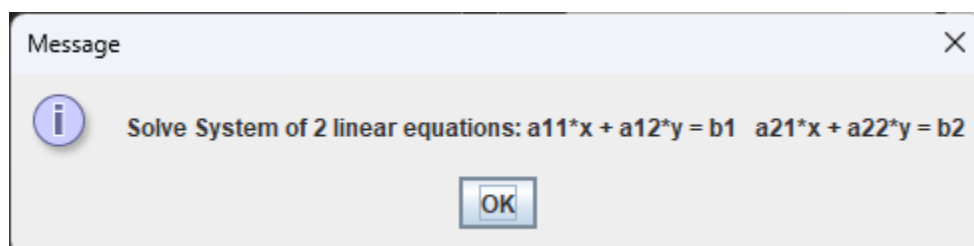
- Result:



+ Solving Linear equation:



+ Solving linear



System of Two equations:

Input

Enter a11:

3

OK Cancel

Input

Enter a12:

4

OK Cancel

Input

Enter b1:

5

OK Cancel

Input

Enter a21:

6

OK Cancel

Input

Enter a22:

7

OK Cancel

Input

Enter b2:

8

OK Cancel

Message

 Solution: $x = -1.0, y = 2.0$

OK

+ Solving Quadratic equation:

Message

 Solve Quadratic Equation: $a^2x + bx + c = 0$

OK

Input

Enter a:

1

OK Cancel

Input

Enter b:

4

OK Cancel

Input

Enter c:

3

OK Cancel

Message

 Two distinct roots:
 $x_1 = -1.0$
 $x_2 = -3.0$

OK

6.1

- Code:

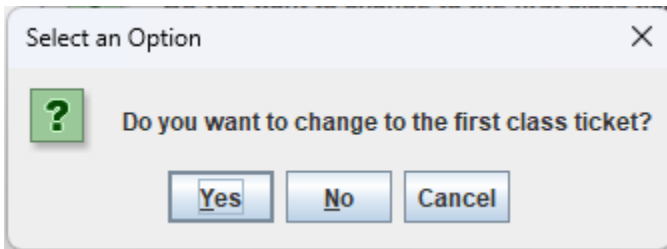
```
package lab01;
import javax.swing.JOptionPane;

public class ChoosingOption {
    public static void main(String[] args) {
        int option = JOptionPane.showConfirmDialog(null,
            "Do you want to change to the first class ticket?");

        JOptionPane.showMessageDialog(null, "You've chosen: "
            + (option == JOptionPane.YES_OPTION ? "Yes" : "No"));

        System.exit(0);
    }
}
```

- Result:



6.2

- Code:

```
package lab01;
import java.util.Scanner;

public class InputFromKeyboard {
    public static void main(String args[]) {
        Scanner keyboard = new Scanner(System.in);

        System.out.println("What's your name?");
        String strName = keyboard.nextLine();
        System.out.println("How old are you?");
        int iAge = keyboard.nextInt();
        System.out.println("How tall are you (m)?");
        double dHeight = keyboard.nextDouble();

        System.out.println("Mrs./Ms. " + strName + ", " + iAge + " years old. "
            + "Your height is " + dHeight + ".");
    }
}
```

- Result:

```
What's your name?
Tuan
How old are you?
19
How tall are you (m)?
1.73
Mrs./Ms. Tuan, 19 years old. Your height is 1.73.
```

6.3

- Code:

```
package lab01;
import java.util.Scanner;
public class star_triangle {
    public static void main(String[] args) {
        Scanner keyboard = new Scanner(System.in);
        System.out.println("Enter height: ");
        int n = keyboard.nextInt();
        int m = 2*n + 1;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j<=n-i; j++) {
                System.out.print(" ");
            }
            for (int j = 1; j<=2*i-1; j++) {
                System.out.print("*");
            }
            for (int j = n+i; j<=m-1; j++) {
                System.out.print(" ");
            }
            System.out.println();
        }
    }
}
```

- Result:

```
Enter height:
5
  *
 ***
*****
*****
*****
```

6.4

- Code:

```
package lab01;
import java.util.Scanner;

public class DaysOfMonth {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int year, month = -1;

        while (true) {
            System.out.print("Enter a valid year: ");
            if (sc.hasNextInt()) {
                year = sc.nextInt();
                if (year >= 0) {
                    break;
                }
            } else {
                sc.next();
            }
            System.out.println("Invalid year! Please enter a valid year (non-negative)");
        }
        sc.nextLine();
        while (month == -1) {
            System.out.print("Enter a valid month: ");
            String input = sc.nextLine().trim().toLowerCase();
            month = convert(input);
            if (month == -1) {
                System.out.println("Invalid month! Please enter a valid month (number 1-12, full name, or abbreviation).");
            }
        }

        int days = getDays(month, year);
        System.out.println("There are " + days + " days in your input month.");
    }
}
```

```

private static int convert(String input) {
    switch (input) {
        case "1": case "january": case "jan.": case "jan": return 1;
        case "2": case "february": case "feb.": case "feb": return 2;
        case "3": case "march": case "mar.": case "mar": return 3;
        case "4": case "april": case "apr.": case "apr": return 4;
        case "5": case "may": return 5;
        case "6": case "june": case "jun.": case "jun": return 6;
        case "7": case "july": case "jul.": case "jul": return 7;
        case "8": case "august": case "aug.": case "aug": return 8;
        case "9": case "september": case "sep.": case "sep": return 9;
        case "10": case "october": case "oct.": case "oct": return 10;
        case "11": case "november": case "nov.": case "nov": return 11;
        case "12": case "december": case "dec.": case "dec": return 12;
        default: return -1;
    }
}

private static int getDays(int month, int year) {
    switch (month) {
        case 1: case 3: case 5: case 7: case 8: case 10: case 12:
            return 31;
        case 4: case 6: case 9: case 11:
            return 30;
        case 2:
            return isLeapYear(year) ? 29 : 28;
    }
    return -1;
}

private static boolean isLeapYear(int year) {
    return (year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);
}
}

```

- Result:

```

Enter a valid year: 2025
Enter a valid month: 10
There are 31 days in your input month.

```

```

Enter a valid year: 2024
Enter a valid month: 2
There are 29 days in your input month.

```

6.5

- Code:

```

package lab01;
import java.util.Arrays;
import java.util.Scanner;

public class SortArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();

        int[] array = new int[n];

        for (int i = 0; i < n; i++) {
            System.out.print("Enter element " + (i + 1) + ": ");
            array[i] = sc.nextInt();
        }

        System.out.println("Original array: " + Arrays.toString(array));

        Arrays.sort(array);
        System.out.println("Sorted array: " + Arrays.toString(array));

        int sum = 0;
        for (int num : array) {
            sum += num;
        }

        double average = (double) sum / array.length;

        System.out.println("Sum = " + sum);
        System.out.println("Average = " + average);
    }
}

```

- Result:

```
Enter number of elements: 5
Enter element 1: 1789
Enter element 2: 2035
Enter element 3: 1899
Enter element 4: 1456
Enter element 5: 2013
Original array: [1789, 2035, 1899, 1456, 2013]
Sorted array: [1456, 1789, 1899, 2013, 2035]
Sum = 9192
Average = 1838.4
```

6.6

- Code:

```
package lab01;
import java.util.Scanner;

public class MatrixSum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();
        System.out.print("Enter number of columns: ");
        int cols = sc.nextInt();

        int[][] A = new int[rows][cols];
        int[][] B = new int[rows][cols];
        int[][] C = new int[rows][cols];

        System.out.println("Enter matrix A:");
        for (int i = 0; i < rows; i++)
            for (int j = 0; j < cols; j++)
                A[i][j] = sc.nextInt();

        System.out.println("Enter matrix B:");
        for (int i = 0; i < rows; i++)
            for (int j = 0; j < cols; j++)
                B[i][j] = sc.nextInt();

        for (int i = 0; i < rows; i++)
            for (int j = 0; j < cols; j++)
                C[i][j] = A[i][j] + B[i][j];

        System.out.println("Result matrix:");
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++)
                System.out.print(C[i][j] + " ");
            System.out.println();
        }
    }
}
```

- Result:

```
Enter number of rows: 3
Enter number of columns: 3
Enter matrix A:
3 4 5
6 7 8
9 10 11
Enter matrix B:
1 3 5
7 9 11
13 15 17
```

```
Result matrix:
4 7 10
13 16 19
22 25 28
```